PROBLEM4:

PROBLEM STATEMENT:

Warehouse Inventory Reconciliation: Compare opening stock, incoming quantities, dispatched quantities, and closing stock across multiple products; detect discrepancies and calculate replenishment requirements.

OBJECTIVE:

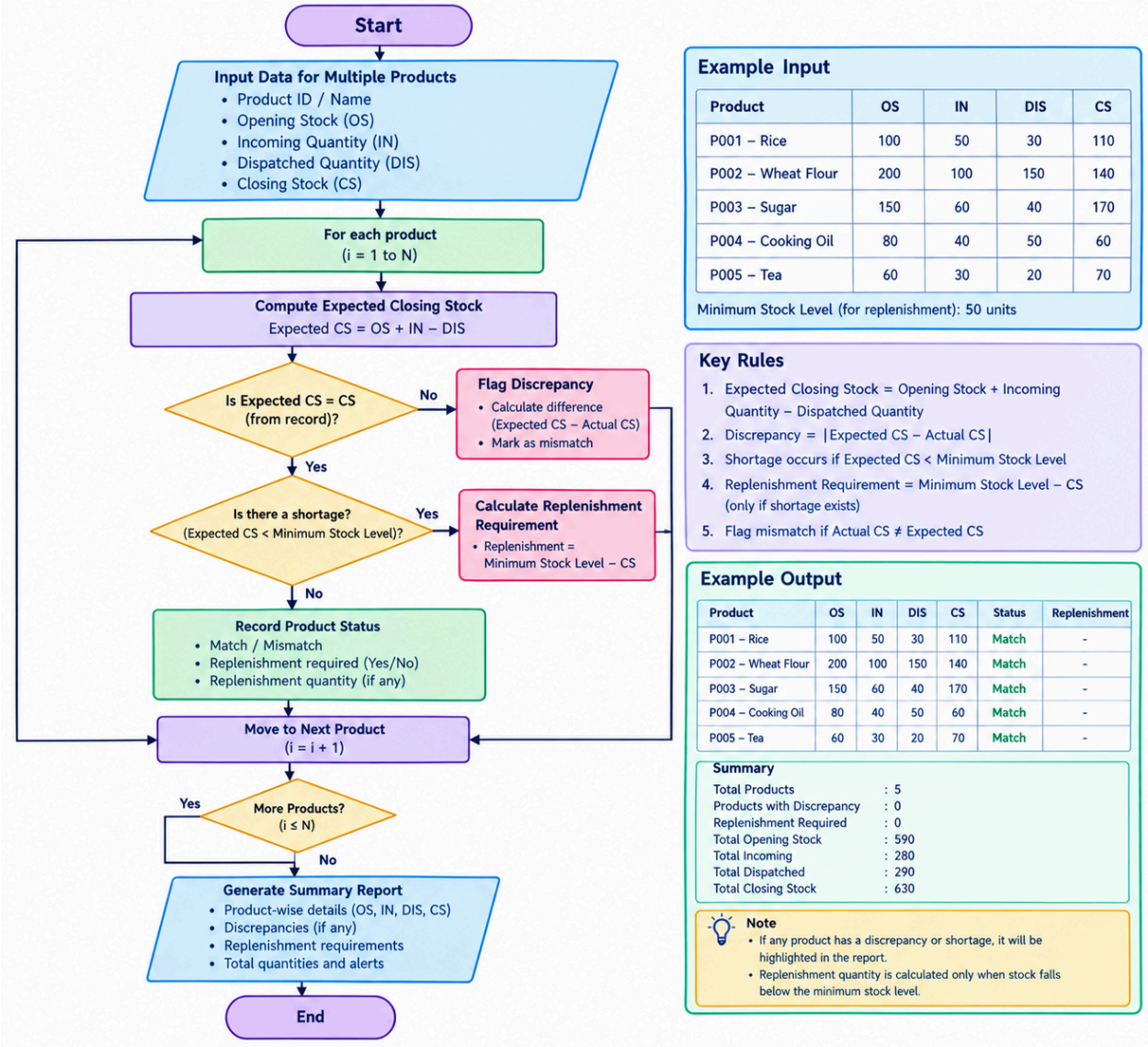
To develop a program that reconciles warehouse inventory for multiple products by comparing opening stock, incoming quantities, dispatched quantities, and actual closing stock. The program detects stock discrepancies, calculates expected closing stock, and determines replenishment requirements for products that fall below the specified minimum stock level.

ALGORITHM:

1. Start.
2. Read the number of products.
3. For each product, input:
    - Product name or ID
    - Opening stock
    - Incoming quantity
    - Dispatched quantity
    - Actual closing stock
    - Minimum stock level
4. Calculate expected closing stock:
   $$
   \text{Expected Closing Stock}
   = \text{Opening Stock}+\text{Incoming Stock}-\text{Dispatched Stock}
   $$
5. Compare expected closing stock with actual closing stock.
6. If both values are equal, mark the inventory as Matched; otherwise, mark it as Discrepancy Detected.
7. Calculate the discrepancy:
   $$
   \text{Discrepancy}=\text{Actual Stock}-\text{Expected Stock}
   $$

8. Calculate replenishment quantity:
   \[
   \text{Replenishment}=\max(0,\text{Minimum Stock}-\text{Actual Stock})
   \]
9. Display the product-wise inventory report.
10. Stop.

FLOWCHART:



PSEUDOCODE:

BEGIN

```
  INPUT n

  FOR i = 1 TO n
    INPUT product_name
    INPUT opening_stock
    INPUT incoming_stock
    INPUT dispatched_stock
    INPUT actual_closing_stock
    INPUT minimum_stock

    expected_stock = opening_stock
                   + incoming_stock
                   - dispatched_stock

    discrepancy = actual_closing_stock
                - expected_stock

    IF discrepancy = 0 THEN
      status = "Matched"
    ELSE
      status = "Discrepancy Detected"
    ENDIF

    replenishment = minimum_stock
                  - actual_closing_stock

    IF replenishment < 0 THEN
      replenishment = 0
    ENDIF

    DISPLAY product_name
    DISPLAY expected_stock
    DISPLAY actual_closing_stock
    DISPLAY discrepancy
    DISPLAY status
    DISPLAY replenishment
  ENDFOR

END
```

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
    int n, i;
    char product[50];
    int opening, incoming, dispatched;
    int actual, minimum;
    int expected, discrepancy, replenishment;

    printf("Enter number of products: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid number of products.\n");
        return 0;
    }

    for (i = 1; i <= n; i++) {
        printf("\nProduct %d\n", i);

        printf("Enter product name: ");
        scanf("%49s", product);

        printf("Enter opening stock: ");
        scanf("%d", &opening);

        printf("Enter incoming stock: ");
        scanf("%d", &incoming);

        printf("Enter dispatched stock: ");
        scanf("%d", &dispatched);

        printf("Enter actual closing stock: ");
        scanf("%d", &actual);

        printf("Enter minimum stock level: ");
        scanf("%d", &minimum);

        if (opening < 0 || incoming < 0 ||
            dispatched < 0 || actual < 0 ||
            minimum < 0 || dispatched > opening + incoming) {
            printf("Invalid stock data.\n");
            continue;
        }
```

```
        expected = opening + incoming - dispatched;
        discrepancy = actual - expected;

        replenishment = minimum - actual;
        if (replenishment < 0) {
            replenishment = 0;
        }

        printf("\nProduct: %s\n", product);
        printf("Expected closing stock: %d\n", expected);
        printf("Actual closing stock: %d\n", actual);
        printf("Discrepancy: %d\n", discrepancy);

        if (discrepancy == 0) {
            printf("Status: Matched\n");
        } else {
            printf("Status: Discrepancy Detected\n");
        }

        printf("Replenishment required: %d\n",
               replenishment);
    }

    return 0;
}
```

OUTPUT:

```
Enter number of products: 3

Product 1
Enter product name: Rice
Enter opening stock: 100
Enter incoming stock: 50
Enter dispatched stock: 80
Enter actual closing stock: 68
Enter minimum stock level: 40

Product: Rice
Expected closing stock: 70
Actual closing stock: 68
Discrepancy: -2
Status: Discrepancy Detected
```

Replenishment required: 0

Product 2
Enter product name: Sugar
Enter opening stock: 60
Enter incoming stock: 20
Enter dispatched stock: 50
Enter actual closing stock: 30
Enter minimum stock level: 35

Product: Sugar
Expected closing stock: 30
Actual closing stock: 30
Discrepancy: 0
Status: Matched
Replenishment required: 5

Product 3
Enter product name: Oil
Enter opening stock: 40
Enter incoming stock: 10
Enter dispatched stock: 25
Enter actual closing stock: 28
Enter minimum stock level: 30

Product: Oil
Expected closing stock: 25
Actual closing stock: 28
Discrepancy: 3
Status: Discrepancy Detected
Replenishment required: 2